

SYSTEMS APPROACH TO TRAINING AND EDUCATION (SATE) CURRICULUM DESIGN PACKAGE: sUAS COURSE (MCRIG)

UNITED STATES MARINE CORPS

1. EXECUTIVE SUMMARY

This document outlines the doctrinally-sound, multi-phase curriculum design for the Marine Corps Robotics Integration Guide / sUAS Course (MCRIG). Developed in strict accordance with NAVMC 1553.1 and NAVMC 1553.2a (Outcomes-Based Learning), this package represents a complete Systems Approach to Training and Education (SATE) development lifecycle.

The sUAS Course is designed to transition entry-level Marines into proficient, safe, and tactically aggressive sUAS Operators and Robotic Integration Practitioners in the Fleet Marine Force. By centering instruction around key Marine Corps Attributes—specifically **Warfighter**, **Decide/Act/Communicate**, and **Exemplary Character**—the curriculum goes beyond task completion, establishing long-term professional dispositions and ethical warrior identities.

2. PHASE 0: MISSION ANALYSIS SUMMARY

2.1 Core Mission Tasks (T&R; Event Inventory)

The training inventory comprises 13 Training and Readiness (T&R;) events spanning Micro (MCRO), Short Range Tactical (SRT), Medium Range Tactical (MRT), Long Range Tactical (LRT), and Attack Drone (AD) platforms.

Event Code	Event Title	Platform Category	Sustainment	Initial Setting
AD-OPS-1001	Perform Handling and Operations	Attack Drone (AD)	1 Month	FORMAL
SUAS-IQT-1002	Conduct Incidental Operator Qualification Training (SRT)	Short Range Tactical (SRT)	Indefinite (0)	FORMAL
SUAS-IQT-1003	Conduct Incidental Operator Qualification Training (MRT)	Medium Range Tactical (MRT)	Indefinite (0)	FORMAL
SUAS-MCRO-1001	Conduct Basic Heads Up Flight	Micro VTOL (MCRO)	Indefinite (0)	FORMAL
SUAS-MCRO-1002	Conduct Basic Heads Down Flight	Micro VTOL (MCRO)	Indefinite (0)	FORMAL
SUAS-MCRO-1003	Conduct Heads Up Precision Flight	Micro VTOL (MCRO)	Indefinite (0)	FORMAL

Event Code	Event Title	Platform Category	Sustainment	Initial Setting
SUAS-SRT-1001	Conduct Basic GUI Flight	Short Range Tactical (SRT)	Indefinite (0)	FORMAL
SUAS-SRT-1002	Conduct Basic System Hand Controller Flight	Short Range Tactical (SRT)	Indefinite (0)	FORMAL
SUAS-MRT-1001	Conduct Basic Flight	Medium Range Tactical (MRT)	Indefinite (0)	FORMAL
SUAS-MRT-1002	Conduct Flight Mode Familiarization and MO Assisted Flight	Medium Range Tactical (MRT)	Indefinite (0)	FORMAL
SUAS-LRT-1001	Conduct Basic Flight	Long Range Tactical (LRT)	Indefinite (0)	FORMAL
SUAS-LRT-1002	Conduct Flight Mode Familiarization Flight	Long Range Tactical (LRT)	Indefinite (0)	FORMAL
SUAS-LRT-1003	Conduct Target Acquisition Flight	Long Range Tactical (LRT)	Indefinite (0)	FORMAL

2.2 Target Audience Baseline & Prerequisites

Students entering the MCRIG sUAS course must satisfy strict administrative, medical, and technical criteria before commencing flight training.

Entry Category	Required Standard	Doctrinal Reference / Details
Basic Qualification	Basic UAS Qualification Level 1 (BUQ-1)	BUQ-1 JSUASMAN Course per JUMTS criteria
Medical Clearance	Annual flight physical and medical screening	Completed NAVMED 6410/13 in accordance with CNAF M-3710.7 and NAVMED 6410/13
Command Approval	Official Designation Letter	Signed by specified ARC and verified by the SUAS PM
Operational Clearance	Approved spectrum and airspace certifications	Certified Restricted Operations Zone (ROZ), approved spectrum request, and signed risk assessment worksheet
Sustainment Profile	Continuous readiness expectation	Sustainment interval is '0 Indefinite' for all tactical platforms, requiring ongoing proficiency. Attack Drone (AD) operations require 1-month sustainment.

2.3 Doctrinal Reference Hierarchy

All curriculum elements are mapped exclusively to the following reference publications. These sources of truth govern aircraft assembly, airframe safety, spectrum deconfliction, and tactical deployment. No external or non-doctrinal sources are permitted to override these publications.

- **CJCSI 3255.01:** Joint UAS Minimum Training Standards (JUMTS)
- **CNAF 3710.9:** Training and Operations Manual Group 1 and 2 Unmanned Aircraft Systems

- **MCRP 3-10.3:** SUAS Operations
- **Applicable System TMs:** System Technical Manuals

3. PHASE 1: PROGRAM LEARNING OUTCOMES (PLOs)

In outcomes-based education, PLOs are professional statements of identity that express how a graduate will 'be' upon course completion. Program outcomes specify the intended professional disposition of the learner and clarify how they will embody these qualities in the Fleet Marine Force.

3.1 Professional Statements of Identity

PLO 1: The Warfighter (Tactical & Technical Integration)

Statement of Identity: An sUAS Operator / Robotic Integration Practitioner is a combat-competent tactical asset, deeply grounded in the employment of diverse aerial platforms (from Micro VTOL to Long-Range Tactical and Attack Drones) to deliver immediate, aggressive, and decisive combat advantages to the infantry unit. Embodying an offensive mindset, the operator actively studies and masters the tactical dynamics of maneuver warfare, target acquisition, and aerial reconnaissance to maintain constant pressure on the adversary and protect friendly forces.

Learning Domain Level: Affective Domain – *Characterization by Value* (The Marine integrates sUAS capability as an organic extension of infantry combat power).

Target Disposition: Tactically aggressive, technically proactive, combat-focused, and offensive-minded.

PLO 2: Decide / Act / Communicate (Cognitive Agility & Decisiveness)

Statement of Identity: An sUAS Operator / Robotic Integration Practitioner is a critical thinker who excels in complex and chaotic environments, rapidly processing real-time aerial intelligence to make sound tactical decisions within the Commander's intent. Possessing a strong bias for action, the operator continuously navigates challenging terrain, adjusts payload sensors for optimal target acquisition, and communicates clear, actionable reports to support immediate fire and maneuver, seizing the tactical initiative on available information.

Learning Domain Level: Cognitive Domain – *Evaluation / Synthesis* (The Marine synthesizes raw sensor data and environmental variables into actionable tactical decisions).

Target Disposition: Situationally aware, decisive under stress, clear and rapid communicator, initiative-driven.

PLO 3: Exemplary Character (Professionalism & Regulatory Stewardship)

Statement of Identity: An sUAS Operator / Robotic Integration Practitioner is a disciplined professional entrusted with high-consequence technology, strictly adhering to safety protocols, Joint UAS Minimum Training Standards (JUMTS), airspace coordination (ROZ) directives, and spectrum regulations. Guided by a strong moral compass, the operator ensures the safe and ethical handling of weaponized or surveillance platforms, demonstrating the 'special trust and confidence' reposed in them by the American people and the command.

Learning Domain Level: Affective Domain – *Valuing* (The Marine internalizes and prioritizes regulatory compliance, airspace safety, and ethical payload deployment as non-negotiable operational baselines).

Target Disposition: Self-disciplined, safety-conscious, regulatory-compliant, and highly accountable.

3.2 Attribute Alignment & Doctrinal Justification

Selected Marine Attribute	Course Integration / Mapping	Doctrinal Justification (NAVMC 1553.2a & T&R)
Warfighter	Directly addresses AD-OPS-1001 (Attack Drones), SUAS-LRT-1003 (Target Acquisition), and SUAS-MRT-1001/2 flight profiles.	Translates raw physical flight control skills into an offensive maneuver capability. Graduates do not just 'fly'; they employ the platform to locate, track, and close with the enemy.
Decide / Act / Communicate	Maps to heads-down navigation, payload adjustments, target tracking, and coordinated flight operations across SUAS-MCRO-1002, SUAS-SRT-1001, and SUAS-LRT-1003.	Requires the operator to filter complex, multi-axis video feeds and GCS data, make immediate piloting/targeting adjustments, and communicate coordinates (MGRS to Lat/Long) to the unit.
Exemplary Character	Maps to strict pre/post-flight inspections, battery management, spectrum requests, ROZ compliance, and safe handling of lethal/inert payloads in AD-OPS-1001 and SUAS-LRT-1001.	Operating autonomous systems and attack drones requires absolute discipline. A failure to adhere to spectrum deconfliction or safety checklists can compromise friendly positions or cause fratricide.

4. PHASE 2: CONDUCT LEARNING ANALYSIS

4.1 Full Uncondensed KSA Brainstorming Report

Every KSA represents a strict 'one verb, one object' format to guarantee observable measurement.

Knowledge Domain (K)

- K1:** Identify flight safety regulations.
- K2:** Recall pre-flight checklist sequences.
- K3:** Explain battery thermal management hazards.
- K4:** Identify radio spectrum allocation procedures.
- K5:** Interpret Restricted Operations Zone (ROZ) dimensions.
- K6:** Describe Ground Control Station (GCS) antenna layouts.
- K7:** Identify real-time flight control telemetry values.
- K8:** Recall emergency aircraft recovery procedures.
- K9:** State weather-related wind limit thresholds.
- K10:** Describe camera payload command interfaces.
- K11:** Explain Automatic In-air Replacement (AIR) parameters.
- K12:** State tactical reconnaissance flight patterns.
- K13:** Describe goggle visual input configurations.

Skills Domain (S)

- S1:** Assemble Ground Control Station hardware.
- S2:** Connect physical controller interface cables.

- S3:** Calibrate Aerial Vehicle (AV) compass sensors.
- S4:** Program autonomous mission flight coordinates.
- S5:** Verify Global Positioning System (GPS) locks.
- S6:** Launch unmanned aerial platforms.
- S7:** Maintain aircraft hover altitude stability.
- S8:** Execute manual flight-control stick inputs.
- S9:** Perform automated flight turn sequences.
- S10:** Direct sensor payload gimbal angles.
- S11:** Identify stationary tactical target signatures.
- S12:** Track moving tactical target signatures.
- S13:** Adjust optical sensor focal zoom.
- S14:** Activate thermal imaging sensor views.
- S15:** Initiate simulated kinetic attack runs.
- S16:** Release inert training target payloads.
- S17:** Coordinate multi-AV physical spacing distances.
- S18:** Handover frequency signal control links.
- S19:** Perform emergency manual landing operations.
- S20:** Execute automated flight return sequences.
- S21:** Execute hand-grab aircraft landing recoveries.
- S22:** Disable vehicle propulsion electric motors.
- S23:** Disassemble Ground Control Station components.
- S24:** Record operator flight log data.

Attitudes Domain (A)

- A1:** Enforce safety checklist procedural discipline.
- A2:** Maintain continuous airspace visual observation.
- A3:** Respect signal spectrum coordination clearances.
- A4:** Prioritize ground crew physical safety.
- A5:** Value accurate system maintenance recording.

4.2 KSA to Draft Behavior Statement (DBS) Mapping Report

Draft Behavior Statement (DBS)	Mapped Knowledge (K) Codes	Mapped Skills (S) Codes	Mapped Attitudes (A) Codes
DBS 1: Regulatory and Airspace Compliance	K1, K4, K5	S24	A3, A4
DBS 2: Ground Control Station Assembly	K6, K13	S1, S2	A1
DBS 3: Mission Planning and Initialization	K2, K9	S3, S4, S5	A1
DBS 4: Launch and Flight Control Dynamics	K7	S6, S7, S8, S9	A2
DBS 5: Sensor Operations and Target Tracking	K10	S10, S11, S12, S13, S14	A2
DBS 6: Kinetic Payload and Attack Runs	K1	S15, S16	A4
DBS 7: Multi-Platform and Control Hand-Offs	K11, K12	S17, S18	A2
DBS 8: Emergency, Recovery, and Storage	K3, K8	S19, S20, S21, S22, S23	A5

4.3 Summarized LAW (Sequenced DBS)

Sequence #	DBS Title	Doctrinal Justification for Sequencing
1	DBS 1: Regulatory and Airspace Compliance	Administrative, airspace (ROZ), and frequency coordination must be mastered prior to handling any real equipment.
2	DBS 2: Ground Control Station Assembly	Students must learn to assemble, power, and configure hardware on the ground before initiating flight setup.
3	DBS 3: Mission Planning and Initialization	Covers checklist execution, calibrations, and loading mission maps prior to launching the aircraft.
4	DBS 4: Launch and Flight Control Dynamics	Teaches the physical mechanics of launching, maneuvering, and maintaining stable flight in multiple profiles.
5	DBS 5: Sensor Operations and Target Tracking	Once stable flight is achieved, students learn to manipulate camera payloads to spot and track targets of interest.
6	DBS 6: Kinetic Payload and Attack Runs	Represents advanced tactical employment of Attack Drones (AD) and dropping payloads on acquired targets.
7	DBS 7: Multi-Platform and Control Hand-Offs	The most complex flight envelope, requiring flight management of multiple aircraft and frequency transitions.
8	DBS 8: Emergency, Recovery, and Storage	Teaches how to handle system anomalies, recover the AV safely (including manual, auto, and air-grabs), and post-flight storage.

4.4 Granular KSA Mapping by T&R Performance Steps (Detailed Event Breakdowns)

Below, selected key representative T&R events are fully decomposed at the performance-step level, using the format specified in the user guidelines.

AD-OPS-1001: Perform Handling and Operations

INITIAL LEARNING SETTING: FORMAL

CONDITION: Given an attack drone, controller, and accessories.

STANDARD: In accordance with safety protocols and proper system operation procedures.

PS 1: Handle the system safely ensuring all safety measures are adhered to prior to operation.

K: Define drone safety protocols.

K: Identify protective equipment requirements.

S: Inspect safety gear functionality.

S: Clear the operating area.

A: Value safe handling practices.

PS 2: Perform pre-flight checks to ensure the battery, motors, sensors, and controls are in operational condition.

K: Describe pre-flight inspection criteria.

K: Recognize battery damage signatures.

S: Measure battery cell voltages.

S: Verify control surface responses.

A: Enforce rigorous check procedures.

PS 3: Set up the goggles and controller, ensuring clear visual and control inputs.

K: Explain video transmitter frequency matching.

K: Identify controller button layouts.

S: Configure goggle video channels.

S: Test stick input latency.

A: Prioritize optimal signal clarity.

PS 4: Perform takeoff using the controller, maintaining altitude and stability.

K: Describe aerodynamic takeoff dynamics.

K: State wind velocity thresholds.

S: Execute throttle stick inputs.

S: Adjust trim levels manually.

A: Maintain calm physical control.

PS 5: Conduct basic flight maneuvers (hover, left/right turns, elevation adjustments) to gain familiarity with the flight dynamics.

K: Describe quad-rotor axes of movement.

S: Control stable stationary hovers.

S: Execute coordinated lateral turns.

S: Perform precision altitude adjustments.

A: Appreciate aircraft flight boundaries.

PS 6: Navigate to a designated area while maintaining constant situational awareness of the environment and obstacles.

- K: Identify tactical airspace boundary markers.
- S: Map environmental obstacles visually.
- S: Navigate designated waypoint sectors.
- A: Maintain continuous visual scanning.

PS 7: Conduct simulated attack runs (using practice payload or simulated effect) on a stationary target, maintaining consistent speed and control.

- K: Describe simulated dive profile angles.
- S: Align drone flight path with target vectors.
- S: Maintain stable attack run speeds.
- A: Maintain focused offensive mindset.

PS 8: Adjust camera settings for optimal target acquisition, ensuring clear visual representation of the target.

- K: Explain digital video zoom parameters.
- K: Identify thermal camera palette options.
- S: Adjust optical camera focal lengths.
- S: Configure digital image exposure settings.
- A: Value high-fidelity target clarity.

PS 9: Deploy payload (if applicable) following proper procedures for accuracy and effectiveness.

- K: Describe payload release mechanisms.
- S: Trigger physical release switches.
- S: Calculate manual drop delay intervals.
- A: Commit to precise release procedures.

PS 10: Perform a safe landing (if applicable) procedure, ensuring the SRL is brought down smoothly and safely after the mission.

- K: Define safe landing area criteria.
- S: Reduce throttle levels progressively.
- S: Align aircraft with landing markers.
- A: Maintain safety during recovery phase.

PS 11: Perform post-flight checks including battery management, drone health inspection, and equipment care.

- K: Identify post-flight damage types.
- K: Describe lithium-polymer storage voltages.
- S: Inspect structural motor mount integrity.
- S: Discharge batteries to storage charge.
- A: Practice diligent equipment preservation.

PS 12: Store the AD and accessories safely to avoid damage or wear on components.

- K: State temperature storage limits.
- S: Package components in protective cases.
- A: Value secure equipment accountability.

SUAS-MCRO-1003: Conduct Heads Up Precision Flight**INITIAL LEARNING SETTING:** FORMAL

CONDITION: Given a Micro Vertical Takeoff and Landing (MCRO) aerial vehicle platform, higher headquarters directives, commander's guidance, and mission.

STANDARD: To perform basic aerial movements, recover the AV, and render it safe without causing injury or damage to personnel or equipment.

PS 1: Setup the GCS.

- K: Explain GCS tablet antenna configurations.
- S: Fasten receiver antennas onto tablet frames.
- A: Prioritize high-quality video links.

PS 2: Assemble and stage the AV.

- K: Identify proper landing gear latch joints.
- S: Click landing gears onto frames securely.
- A: Value structural flight durability checks.

PS 3: Conduct preflight checks and prepare for launch.

- K: Recall local wind speed flight limits.
- S: Gauge wind levels using anemometers.
- A: Enforce conservative flight launch choices.

PS 4: Launch the AV, being conscious of obstacles and taking actions to clear them.

- K: Identify low-hanging tactical obstacle hazards.
- S: Guide aircraft off launcher stages cleanly.
- A: Maintain extreme obstacle clearance focus.

PS 5: Fly AV to a specified obstacle or set of obstacles while heads up.

- K: State tactical obstacle safety distances.
- S: Guide aircraft near physical obstacles visually.
- A: Maintain heightened flight path alertness.

PS 6: Maneuver the AV in close proximity to the obstacle or set of obstacles in order to simulate close reconnaissance of specified items / targets of interest while heads up.

- K: Describe vortex-ring aerodynamic hazards near walls.
- S: Execute precision close-proximity hover patterns.
- A: Prioritize platform safety over close tracks.

PS 7: Bring the AV home using the GUI while heads up.

- K: Recall auto return-home height clearance values.
- S: Press automatic GCS home switches.
- A: Rely on backup auto-recovery routes.

PS 8: Bring the AV to the VO to prepare for an air grab, ensuring that the AV is aligned so that the camera is pointed away from the VO.

- K: Explain safe air-grab camera angle orientations.
- S: Hover aircraft in visual hand reach.

S: Rotate cameras outward from ground operators.

A: Prioritize ground operator physical safety.

PS 9: Execute an air grab of the AV and kill the motors while maintaining the AV at arm's length.

K: Identify safe air-grab frame grasping surfaces.

S: Grab active composite airframes firmly.

S: Shut down flight motors instantaneously.

A: Commit to physical safety rules in action.

PS 10: Render the AV safe.

K: State micro-VTOL component storage limits.

S: Pack delicate parts in secure transport cases.

A: Maintain diligent equipment preservation.

5. PHASE 3: LEARNING OBJECTIVES WORKSHEETS (LOWs)

To comply with SATE requirements, individual LOWs are generated below for each of the 13 T&R events. TLOs map verbatim, and ELOs utilize observable verbs with specific qualifiers (such as *in writing*, *orally*, *by manipulating the controls*).

5.1 LOW for AD-OPS-1001: Perform Handling and Operations

Terminal Learning Objective (TLO)

- **Task:** Perform Handling and Operations
- **Condition:** Given an attack drone, controller, and accessories.
- **Standard:** In accordance with safety protocols and proper system operation procedures.

Enabling Learning Objectives (ELOs)

- **AD-OPS-1001-ELO-1 (Cognitive):** *Identify, in writing on a written exam*, core drone safety protocols and protective gear requirements. Standard: Without error, achieving a minimum score of 80%.
- **AD-OPS-1001-ELO-2 (Cognitive):** *Explain orally to an evaluator*, battery thermal management hazards and storage threshold parameters. Standard: In accordance with the applicable System Technical Manual (TM).
- **AD-OPS-1001-ELO-3 (Psychomotor):** *Perform, by manipulating the physical hand controller*, basic flight maneuvers including hover, coordinated turns, and elevation adjustments. Standard: Safely maintaining altitude within +/- 2 meters of the target hover point.
- **AD-OPS-1001-ELO-4 (Psychomotor):** *Conduct, by executing GCS touchscreen inputs*, a simulated attack run on a designated stationary target. Standard: Maintaining a consistent dive speed and tracking target center throughout the run.
- **AD-OPS-1001-ELO-5 (Affective):** *Enforce, by physically verifying the pre-flight safety checklist*, proper safety protocols prior to aircraft launch. Standard: Ensuring zero injury to personnel or damage to equipment.

5.2 LOW for SUAS-IQT-1002: Conduct Incidental Operator Qualification Training (SRT)

Terminal Learning Objective (TLO)

- **Task:** Conduct Incidental Operator Qualification Training (SRT)
- **Condition:** Given a Short Range Tactical (SRT) aerial vehicle platform, higher headquarters directives, commander's guidance, and mission.
- **Standard:** To support mission requirements and assess the results of the action, and issues subsequent orders as required.

Enabling Learning Objectives (ELOs)

- **SUAS-IQT-1002-ELO-1 (Cognitive):** *Describe, in writing on a diagnostic sheet, the operational steps of an Automatic In-air Replacement (AIR) sequence.* Standard: Verbatim matching the sequence outlined in CJCSI 3255.01.
- **SUAS-IQT-1002-ELO-2 (Psychomotor):** *Execute, by manipulating the GCS stylus controls, a basic GUI flight profile using coordinate waypoint drops.* Standard: Accurately plotting and passing through all designated map coordinates.
- **SUAS-IQT-1002-ELO-3 (Psychomotor):** *Perform, by adjusting hand-controller joystick inputs, daytime target acquisition and offset target tracking flights.* Standard: Keeping the targeted vehicle centered in the camera feed for a minimum of 2 continuous minutes.
- **SUAS-IQT-1002-ELO-4 (Psychomotor):** *Conduct, by manipulating hand-controller dials, night flight operations utilizing thermal camera sensor feeds.* Standard: Safely navigating and identifying simulated enemy targets under zero-ambient-light conditions.
- **SUAS-IQT-1002-ELO-5 (Affective):** *Prioritize, by executing standard deconfliction radio calls, safe airspace separation limits during multi-UAV flight operations.* Standard: Ensuring no near-miss events or unauthorized airspace penetrations occur.

5.3 LOW for SUAS-IQT-1003: Conduct Incidental Operator Qualification Training (MRT)

Terminal Learning Objective (TLO)

- **Task:** Conduct Incidental Operator Qualification Training (MRT)
- **Condition:** Given a Medium Range Tactical (MRT) aerial vehicle platform, higher headquarters directives, commander's guidance, and mission.
- **Standard:** To support mission requirements and assess the results of the action, and issues subsequent orders as required.

Enabling Learning Objectives (ELOs)

- **SUAS-IQT-1003-ELO-1 (Cognitive):** *Explain orally to the flight instructor, the aerodynamic differences and hazards associated with hand-launched vs. catapult-launched flight modes.* Standard: In accordance with CNAF 3710.9 operations guidelines.
- **SUAS-IQT-1003-ELO-2 (Psychomotor):** *Demonstrate, by manipulating GCS menu commands, a digital vehicle control handover (AV Hand-Off) from GCS-1 to GCS-2.* Standard: Successfully completing the signal transfer in under 30 seconds without loss of aircraft control.
- **SUAS-IQT-1003-ELO-3 (Psychomotor):** *Conduct, by using GCS telemetry menus and physical controls, a mobile flight operation while operating from a moving vehicle.* Standard: Maintaining stable communication links and

executing a safe recovery.

- **SUAS-IQT-1003-ELO-4 (Psychomotor):** *Pilot, by navigating manual payload camera configurations, the aircraft under low-level obstacles (LL flight) to land at a designated point. Standard: Preventing collision and completing touchdown within 1 meter of the marked zone.*
- **SUAS-IQT-1003-ELO-5 (Affective):** *Enforce, by utilizing clear verbal directions, exact safety parameters when conducting flight operations with the aid of an untrained assistant. Standard: Guaranteeing the safety of the assistant throughout launching and recovery sequences.*

5.4 LOW for SUAS-MCRO-1001: Conduct Basic Heads Up Flight

Terminal Learning Objective (TLO)

- **Task:** Conduct Basic Heads Up Flight
- **Condition:** Given a Micro Vertical Takeoff and Landing (MCRO) aerial vehicle platform, higher headquarters directives, commander's guidance, and mission.
- **Standard:** To perform basic aerial movements, recover the AV, and render it safe without causing injury or damage to personnel or equipment.

Enabling Learning Objectives (ELOs)

- **SUAS-MCRO-1001-ELO-1 (Cognitive):** *Recall, in writing on a pre-flight checklist, the minimum GPS satellite lock count required for safe autonomous micro-VTOL launching. Standard: Expressed as a minimum of 8 active satellite connections.*
- **SUAS-MCRO-1001-ELO-2 (Psychomotor):** *Execute, by manipulating physical controller sticks in visual line of sight, flat box patterns utilizing only lateral control inputs. Standard: Maintaining a consistent visual height of 10 feet AGL +/- 0.5 meters.*
- **SUAS-MCRO-1001-ELO-3 (Psychomotor):** *Demonstrate, by blending lateral and yaw controller stick movements, a continuous-action box pattern with coordinated turns at the corners. Standard: Executing a fluid, non-stop flight path around four designated ground markers.*
- **SUAS-MCRO-1001-ELO-4 (Psychomotor):** *Perform, by selecting touch commands on the tablet screen, an autonomous return-to-home and landing sequence. Standard: Bringing the aircraft down to a safe, controlled stop within the designated landing zone.*
- **SUAS-MCRO-1001-ELO-5 (Affective):** *Value, by conducting systematic physical battery connections and checks, micro hardware component integrity prior to and after flight. Standard: Ensuring no electrical short-circuits or connection failures occur during operation.*

5.5 LOW for SUAS-MCRO-1002: Conduct Basic Heads Down Flight

Terminal Learning Objective (TLO)

- **Task:** Conduct Basic Heads Down Flight
- **Condition:** Given a Micro Vertical Takeoff and Landing (MCRO) aerial vehicle platform, higher headquarters directives, commander's guidance, and mission.
- **Standard:** To perform basic aerial movements, recover the AV, and render it safe without causing injury or damage to personnel or equipment.

Enabling Learning Objectives (ELOs)

- **SUAS-MCRO-1002-ELO-1 (Cognitive):** *State orally to an evaluator, the optical lookup angles and GCS menu sub-items used to navigate around obstacles during blind flight.* Standard: Aligning with the system technical manual configurations.
- **SUAS-MCRO-1002-ELO-2 (Psychomotor):** *Perform, by using on-screen GCS telemetry instruments and video feeds exclusively, multi-axis flight maneuvers and box pattern tracks.* Standard: Maintaining constant altitude and obstacle clearance without looking directly at the aircraft.
- **SUAS-MCRO-1002-ELO-3 (Psychomotor):** *Manipulate, by adjusting on-screen gimbal settings, camera payload look-angles to locate and identify target silhouettes.* Standard: Accurately identifying 100% of simulated targets within a 200m radius.
- **SUAS-MCRO-1002-ELO-4 (Psychomotor):** *Land, by using manual GUI inputs on the GCS screen, the aircraft at an offset safe location at least 10 meters away from the home base.* Standard: Achieving a smooth touchdown without bouncing or tips.
- **SUAS-MCRO-1002-ELO-5 (Affective):** *Practice, by executing immediate climb procedures upon detection of signal fade, active RF link monitoring during deep-altitude climbs.* Standard: Preventing signal loss or loss-of-control events.

5.6 LOW for SUAS-MCRO-1003: Conduct Heads Up Precision Flight

Terminal Learning Objective (TLO)

- **Task:** Conduct Heads Up Precision Flight
- **Condition:** Given a Micro Vertical Takeoff and Landing (MCRO) aerial vehicle platform, higher headquarters directives, commander's guidance, and mission.
- **Standard:** To perform basic aerial movements, recover the AV, and render it safe without causing injury or damage to personnel or equipment.

Enabling Learning Objectives (ELOs)

- **SUAS-MCRO-1003-ELO-1 (Cognitive):** *Identify, in writing on a written diagram, the designated safe structural hand-grasping zones of a micro-VTOL airframe.* Standard: Highlighting only approved areas to avoid motor blades and sensors.
- **SUAS-MCRO-1003-ELO-2 (Psychomotor):** *Maneuver, by utilizing fine hand-stick movements in visual flight, the aircraft in close proximity to physical obstacles to capture target details.* Standard: Maintaining a consistent standoff distance between 1 and 3 meters from the obstacle.
- **SUAS-MCRO-1003-ELO-3 (Psychomotor):** *Position, by manipulating flight control inputs, the aircraft into an air-grab hover adjacent to the Visual Observer (VO) with the camera lens pointed away from the operator.* Standard: Holding a steady hover within 1.5 meters of the VO's reaching position.
- **SUAS-MCRO-1003-ELO-4 (Psychomotor):** *Execute, by physically grabbing the airframe and immediately killing the propulsion system, a safe hand-recovery (air-grab) of the aircraft.* Standard: Grabbing the frame at arm's length and stopping motors cleanly without injury.
- **SUAS-MCRO-1003-ELO-5 (Affective):** *Prioritize, by executing immediate motor-kill stick inputs in emergency hand-grabs, ground operator safety during recovery sequences.* Standard: Eliminating any possibility of skin contact with rotating propeller blades.

5.7 LOW for SUAS-SRT-1001: Conduct Basic GUI Flight

Terminal Learning Objective (TLO)

- **Task:** Conduct Basic GUI Flight
- **Condition:** Given a Short Range Tactical (SRT) aerial vehicle platform, higher headquarters directives, commander's guidance, and mission.
- **Standard:** To perform basic aerial movements, recover the AV, and render it safe without causing injury or damage to personnel or equipment.

Enabling Learning Objectives (ELOs)

- **SUAS-SRT-1001-ELO-1 (Cognitive):** *State orally to an evaluator, the steps to pair GCS systems to specific electronic serial numbers of SRT platforms.* Standard: In accordance with the GCS operation handbook procedures.
- **SUAS-SRT-1001-ELO-2 (Psychomotor):** *Assemble, by connecting antenna masts and GCS components physically, the baseline ground GCS station.* Standard: Completing setup within a 5-minute time limit.
- **SUAS-SRT-1001-ELO-3 (Psychomotor):** *Perform, by touching the GCS stylus onto the map interface, real-time coordinate waypoint placements and automated flight routing.* Standard: Creating an active flight plan of at least three points with immediate execution.
- **SUAS-SRT-1001-ELO-4 (Psychomotor):** *Command, by pushing the physical RTH button on the GCS panel, an automated Return-to-Home and autonomous terrain landing.* Standard: Executing touchdown smoothly on a cleared surface.
- **SUAS-SRT-1001-ELO-5 (Affective):** *Enforce, by conducting physical pre-flight check sweeps, visual airspace clearing procedures before starting automated launches.* Standard: Ensuring zero local airspace conflicts occur during launch.

5.8 LOW for SUAS-SRT-1002: Conduct Basic System Hand Controller Flight

Terminal Learning Objective (TLO)

- **Task:** Conduct Basic System Hand Controller Flight
- **Condition:** Given a Short Range Tactical (SRT) aerial vehicle platform, higher headquarters directives, commander's guidance, and mission.
- **Standard:** To perform basic aerial movements, recover the AV, and render it safe without causing injury or damage to personnel or equipment.

Enabling Learning Objectives (ELOs)

- **SUAS-SRT-1002-ELO-1 (Cognitive):** *Identify, in writing on a layout sheet, physical controller button functions and analog joystick spring-tension behaviors.* Standard: Scoring 100% on the button mapping template.
- **SUAS-SRT-1002-ELO-2 (Psychomotor):** *Execute, by utilizing physical controller joysticks during visually-observed flight, flat box pattern maneuvers.* Standard: Keeping the aircraft stable with minimal wind-drift deviations.
- **SUAS-SRT-1002-ELO-3 (Psychomotor):** *Execute, by utilizing on-screen video feeds and physical joysticks, a coordinated banking corner turn in a box flight pattern.* Standard: Maintaining course lines with minimal overshoot errors.

- **SUAS-SRT-1002-ELO-4 (Psychomotor):** *Perform, by manipulating the physical slider dial on the hand controller, real-time drone speed changes and gimbal camera pitches. Standard: Smoothly adjusting speed and targeting views without jerking the aircraft.*
- **SUAS-SRT-1002-ELO-5 (Affective):** *Practice, by executing immediate dual-stick emergency motor kill commands upon touchdown, precise propulsion deactivation. Standard: Preventing prop wash and rollover damage on landing.*

5.9 LOW for SUAS-MRT-1001: Conduct Basic Flight

Terminal Learning Objective (TLO)

- **Task:** Conduct Basic Flight
- **Condition:** Given a Medium Range Tactical (MRT) aerial vehicle platform, higher headquarters directives, commander's guidance, and mission.
- **Standard:** To perform basic aerial movements, recover the AV, and render it safe without causing injury or damage to personnel or equipment.

Enabling Learning Objectives (ELOs)

- **SUAS-MRT-1001-ELO-1 (Cognitive):** *State, in writing on a technical worksheet, the exact aerodynamic speeds and wind limit boundaries of the MRT platform. Standard: Verbatim matching the system's operational technical specifications.*
- **SUAS-MRT-1001-ELO-2 (Psychomotor):** *Assemble, by locking wing pins and attaching control rods, the airframe and mechanical launcher rail system. Standard: Completing physical assembly within 15 minutes.*
- **SUAS-MRT-1001-ELO-3 (Psychomotor):** *Perform, by using physical switch toggles and software controls on the laptop, flight mode shifts between MAN, ALT, and NAV modes. Standard: Executing mode changes smoothly at designated altitude levels.*
- **SUAS-MRT-1001-ELO-4 (Psychomotor):** *Execute, by manipulating flight commands on the RSTA laptop interface, timed turns, orbits, and a teardrop landing approach. Standard: Keeping the flight track centered along the pre-planned approach path.*
- **SUAS-MRT-1001-ELO-5 (Affective):** *Prioritize, by physically securing launcher safety pins and clearing the rail track, safety boundaries around the catapult launch system. Standard: Preventing physical launcher hazards to ground crew.*

5.10 LOW for SUAS-MRT-1002: Conduct Flight Mode Familiarization and MO Assisted Flight

Terminal Learning Objective (TLO)

- **Task:** Conduct Flight Mode Familiarization and MO Assisted Flight
- **Condition:** Given a Medium Range Tactical (MRT) aerial vehicle platform, higher headquarters directives, commander's guidance, and mission.
- **Standard:** To perform basic aerial movements, recover the AV, and render it safe without causing injury or damage to personnel or equipment.

Enabling Learning Objectives (ELOs)

- **SUAS-MRT-1002-ELO-1 (Cognitive):** *Describe orally to the evaluator, the programmatic differences between MGRS coordinate grid frames and Latitude/Longitude coordinates. Standard: Demonstrating understanding of tactical coordinate conversions in-flight.*
- **SUAS-MRT-1002-ELO-2 (Psychomotor):** *Convert, by selecting software menu options on the RSTA laptop, coordinate formats from MGRS to LAT/LONG and back during active flight. Standard: Correctly executing the toggle and logging target locations in under 10 seconds.*
- **SUAS-MRT-1002-ELO-3 (Psychomotor):** *Modify, by editing coordinates on the Mission Planning Tool live, waypoint positions, altitudes, and orbit rotation directions. Standard: Uploading and synchronizing the new flight plan successfully during autonomous flight.*
- **SUAS-MRT-1002-ELO-4 (Psychomotor):** *Maneuver, by selecting GCS tracking modes, the camera payload to slave the camera focal point to specific terrain coordinates. Standard: Keeping the targeted ground coordinates centered in the video feed.*
- **SUAS-MRT-1002-ELO-5 (Affective):** *Practice, by executing immediate verbal callouts and task-sharing actions, structured crew coordination between the Mission Operator (MO) and pilot. Standard: Completing all flight adjustments inside a smooth, coordinated cockpit environment.*

5.11 LOW for SUAS-LRT-1001: Conduct Basic Flight

Terminal Learning Objective (TLO)

- **Task:** Conduct Basic Flight
- **Condition:** Given a Long Range Tactical (LRT) aerial vehicle platform, higher headquarters directives, commander's guidance, and mission.
- **Standard:** Conduct target acquisition procedures, recover the AV, and render it safe without causing injury or damage to personnel or equipment.

Enabling Learning Objectives (ELOs)

- **SUAS-LRT-1001-ELO-1 (Cognitive):** *Identify, in writing on a written exam, LRT GCS antenna mast safety grounding methods and local airspace boundary laws. Standard: Correctly identifying all safety protocols with 100% accuracy.*
- **SUAS-LRT-1001-ELO-2 (Psychomotor):** *Set up, by raising radio masts and connecting data cables, the LRT Ground Control Station. Standard: Verifying signal-to-noise ratios are within safe operational limits.*
- **SUAS-LRT-1001-ELO-3 (Psychomotor):** *Execute, by utilizing GCS computer keyboard tap shortcuts and U-Drive controls, manual aircraft flight commands and payload adjustments. Standard: Correctly navigating and maintaining the target glide slope.*
- **SUAS-LRT-1001-ELO-4 (Psychomotor):** *Command, by selecting touch actions on the mission laptop, an autonomous descent and automated landing touchdown. Standard: Safely recovering the aircraft on the designated runway.*
- **SUAS-LRT-1001-ELO-5 (Affective):** *Value, by conducting physical pre-flight structural wing lock inspections, complete equipment integrity checks. Standard: Eliminating any risk of structural component failure in-flight.*

5.12 LOW for SUAS-LRT-1002: Conduct Flight Mode Familiarization Flight

Terminal Learning Objective (TLO)

- **Task:** Conduct Flight Mode Familiarization Flight
- **Condition:** Given a Long Range Tactical (LRT) aerial vehicle platform, higher headquarters directives, commander's guidance, and mission.
- **Standard:** Conduct target acquisition procedures, recover the AV, and render it safe without causing injury or damage to personnel or equipment.

Enabling Learning Objectives (ELOs)

- **SUAS-LRT-1002-ELO-1 (Cognitive):** *Explain, in writing on an exam*, the operational functions and conditions for using 'Orbit Here Now' and image-capture commands. Standard: Verbatim matching the system technical manual definitions.
- **SUAS-LRT-1002-ELO-2 (Psychomotor):** *Execute, by clicking on-screen software options*, the 'Orbit Here Now' flight hold command at an active coordinate position. Standard: Holding a circular flight pattern over the target within 10 meters of coordinate center.
- **SUAS-LRT-1002-ELO-3 (Psychomotor):** *Upload, by using the GCS live-sync transmission menu*, modified waypoint sequences to the aircraft during flight. Standard: Completing the data sync without causing flight path errors.
- **SUAS-LRT-1002-ELO-4 (Psychomotor):** *Capture, by using GCS snapshot hotkeys*, high-resolution digital target images of a specified ground point. Standard: Capturing a clear image with no blur or focal distortion.
- **SUAS-LRT-1002-ELO-5 (Affective):** *Support, by following methodical post-flight component teardown steps*, strict hardware handling and storage rules. Standard: Preventing physical damage or electrical wear to sensitive systems.

5.13 LOW for SUAS-LRT-1003: Conduct Target Acquisition Flight

Terminal Learning Objective (TLO)

- **Task:** Conduct Target Acquisition Flight
- **Condition:** Given a Long Range Tactical (LRT) aerial vehicle platform, higher headquarters directives, commander's guidance, and mission.
- **Standard:** Conduct target acquisition procedures, recover the AV, and render it safe without causing injury or damage to personnel or equipment.

Enabling Learning Objectives (ELOs)

- **SUAS-LRT-1003-ELO-1 (Cognitive):** *Identify orally to an evaluator*, the critical visual differences between civilian vehicles and military armor signatures. Standard: Correctly identifying 100% of target silhouettes on a test sheet.
- **SUAS-LRT-1003-ELO-2 (Psychomotor):** *Spot, by scanning the payload video streams during simulated flight*, vehicle silhouettes and items of tactical interest on the terrain. Standard: Identifying all target locations in the operational search area.
- **SUAS-LRT-1003-ELO-3 (Psychomotor):** *Modify, by editing coordinates live on the mission map*, the flight path to orbit directly over a spotted target of interest. Standard: Syncing and executing the revised orbit in under 20 seconds.

- **SUAS-LRT-1003-ELO-4 (Psychomotor):** *Track, by engaging the GCS automatic target tracking software loops, moving and stationary targets using the camera gimbal. Standard: Keeping the target in continuous camera view through multi-axis movements.*
- **SUAS-LRT-1003-ELO-5 (Affective):** *Value, by conducting thorough physical airframe and log inspections post-flight, meticulous safety recordkeeping. Standard: Log data must match actual flight timers within a tolerance of 1 minute.*

6. PHASE 4: DEVELOP COURSE MATERIALS

6.1 Table of Specifications (TOS)

The TOS outlines the exact evaluation blueprint for sUAS curriculum validation, matching ELOs to specific testing modes and item parameters.

Objective #	Domain	Level of Learning	Assessment Method	# of Items
AD-OPS-1001-ELO-1	Cognitive	Knowledge (Recall)	Written Exam	5 Multiple Choice
AD-OPS-1001-ELO-2	Cognitive	Comprehension (Explain)	Oral Examination	1 Essay/Explain Prompt
AD-OPS-1001-ELO-3	Psychomotor	Guided Response (Perform)	Practical Evaluation (PECL)	1 Rubric Step
AD-OPS-1001-ELO-4	Psychomotor	Mechanism (Conduct)	Practical Evaluation (PECL)	1 Rubric Step
AD-OPS-1001-ELO-5	Affective	Valuing (Enforce)	Practical Evaluation (PECL)	1 Rubric Step
SUAS-IQT-1002-ELO-1	Cognitive	Knowledge (Describe)	Written Exam	3 Multiple Choice
SUAS-IQT-1002-ELO-2	Psychomotor	Guided Response (Execute)	Practical Evaluation (PECL)	1 Rubric Step
SUAS-IQT-1002-ELO-3	Psychomotor	Mechanism (Perform)	Practical Evaluation (PECL)	1 Rubric Step
SUAS-IQT-1002-ELO-4	Psychomotor	Mechanism (Conduct)	Practical Evaluation (PECL)	1 Rubric Step
SUAS-IQT-1002-ELO-5	Affective	Valuing (Prioritize)	Practical Evaluation (PECL)	1 Rubric Step
SUAS-IQT-1003-ELO-1	Cognitive	Comprehension (Explain)	Oral Examination	1 Essay/Explain Prompt
SUAS-IQT-1003-ELO-2	Psychomotor	Complex Overt Response	Practical Evaluation (PECL)	1 Rubric Step
SUAS-IQT-1003-ELO-3	Psychomotor	Complex Overt Response	Practical Evaluation (PECL)	1 Rubric Step

Objective #	Domain	Level of Learning	Assessment Method	# of Items
SUAS-IQT-1003-ELO-4	Psychomotor	Complex Overt Response	Practical Evaluation (PECL)	1 Rubric Step
SUAS-IQT-1003-ELO-5	Affective	Organization (Enforce)	Practical Evaluation (PECL)	1 Rubric Step
SUAS-MCRO-1001-ELO-1	Cognitive	Knowledge (Recall)	Written Exam	2 Multiple Choice
SUAS-MCRO-1001-ELO-2	Psychomotor	Guided Response (Execute)	Practical Evaluation (PECL)	1 Rubric Step
SUAS-MCRO-1001-ELO-3	Psychomotor	Mechanism (Demonstrate)	Practical Evaluation (PECL)	1 Rubric Step
SUAS-MCRO-1001-ELO-4	Psychomotor	Guided Response (Perform)	Practical Evaluation (PECL)	1 Rubric Step
SUAS-MCRO-1001-ELO-5	Affective	Valuing (Value)	Practical Evaluation (PECL)	1 Rubric Step
SUAS-MCRO-1002-ELO-1	Cognitive	Knowledge (State)	Oral Examination	1 Essay/Explain Prompt
SUAS-MCRO-1002-ELO-2	Psychomotor	Mechanism (Perform)	Practical Evaluation (PECL)	1 Rubric Step
SUAS-MCRO-1002-ELO-3	Psychomotor	Mechanism (Manipulate)	Practical Evaluation (PECL)	1 Rubric Step
SUAS-MCRO-1002-ELO-4	Psychomotor	Guided Response (Land)	Practical Evaluation (PECL)	1 Rubric Step
SUAS-MCRO-1002-ELO-5	Affective	Characterization (Practice)	Practical Evaluation (PECL)	1 Rubric Step
SUAS-MCRO-1003-ELO-1	Cognitive	Knowledge (Identify)	Written Exam	2 Multiple Choice
SUAS-MCRO-1003-ELO-2	Psychomotor	Mechanism (Maneuver)	Practical Evaluation (PECL)	1 Rubric Step
SUAS-MCRO-1003-ELO-3	Psychomotor	Guided Response (Position)	Practical Evaluation (PECL)	1 Rubric Step
SUAS-MCRO-1003-ELO-4	Psychomotor	Mechanism (Execute)	Practical Evaluation (PECL)	1 Rubric Step
SUAS-MCRO-1003-ELO-5	Affective	Characterization (Prioritize)	Practical Evaluation (PECL)	1 Rubric Step
SUAS-SRT-1001-ELO-1	Cognitive	Knowledge (State)	Oral Examination	1 Essay/Explain Prompt
SUAS-SRT-1001-ELO-2	Psychomotor	Guided Response (Assemble)	Practical Evaluation (PECL)	1 Rubric Step

Objective #	Domain	Level of Learning	Assessment Method	# of Items
SUAS-SRT-1001-ELO-3	Psychomotor	Mechanism (Perform)	Practical Evaluation (PECL)	1 Rubric Step
SUAS-SRT-1001-ELO-4	Psychomotor	Guided Response (Command)	Practical Evaluation (PECL)	1 Rubric Step
SUAS-SRT-1001-ELO-5	Affective	Valuing (Enforce)	Practical Evaluation (PECL)	1 Rubric Step
SUAS-SRT-1002-ELO-1	Cognitive	Knowledge (Identify)	Written Exam	3 Multiple Choice
SUAS-SRT-1002-ELO-2	Psychomotor	Guided Response (Execute)	Practical Evaluation (PECL)	1 Rubric Step
SUAS-SRT-1002-ELO-3	Psychomotor	Mechanism (Execute)	Practical Evaluation (PECL)	1 Rubric Step
SUAS-SRT-1002-ELO-4	Psychomotor	Mechanism (Perform)	Practical Evaluation (PECL)	1 Rubric Step
SUAS-SRT-1002-ELO-5	Affective	Characterization (Practice)	Practical Evaluation (PECL)	1 Rubric Step
SUAS-MRT-1001-ELO-1	Cognitive	Knowledge (State)	Written Exam	4 Multiple Choice
SUAS-MRT-1001-ELO-2	Psychomotor	Guided Response (Assemble)	Practical Evaluation (PECL)	1 Rubric Step
SUAS-MRT-1001-ELO-3	Psychomotor	Mechanism (Perform)	Practical Evaluation (PECL)	1 Rubric Step
SUAS-MRT-1001-ELO-4	Psychomotor	Mechanism (Execute)	Practical Evaluation (PECL)	1 Rubric Step
SUAS-MRT-1001-ELO-5	Affective	Valuing (Prioritize)	Practical Evaluation (PECL)	1 Rubric Step
SUAS-MRT-1002-ELO-1	Cognitive	Comprehension (Describe)	Oral Examination	1 Essay/Explain Prompt
SUAS-MRT-1002-ELO-2	Psychomotor	Mechanism (Convert)	Practical Evaluation (PECL)	1 Rubric Step
SUAS-MRT-1002-ELO-3	Psychomotor	Mechanism (Modify)	Practical Evaluation (PECL)	1 Rubric Step
SUAS-MRT-1002-ELO-4	Psychomotor	Mechanism (Maneuver)	Practical Evaluation (PECL)	1 Rubric Step
SUAS-MRT-1002-ELO-5	Affective	Organization (Practice)	Practical Evaluation (PECL)	1 Rubric Step
SUAS-LRT-1001-ELO-1	Cognitive	Knowledge (Identify)	Written Exam	3 Multiple Choice

Objective #	Domain	Level of Learning	Assessment Method	# of Items
SUAS-LRT-1001-ELO-2	Psychomotor	Guided Response (Set up)	Practical Evaluation (PECL)	1 Rubric Step
SUAS-LRT-1001-ELO-3	Psychomotor	Mechanism (Execute)	Practical Evaluation (PECL)	1 Rubric Step
SUAS-LRT-1001-ELO-4	Psychomotor	Guided Response (Command)	Practical Evaluation (PECL)	1 Rubric Step
SUAS-LRT-1001-ELO-5	Affective	Valuing (Value)	Practical Evaluation (PECL)	1 Rubric Step
SUAS-LRT-1002-ELO-1	Cognitive	Knowledge (Explain)	Written Exam	4 Multiple Choice
SUAS-LRT-1002-ELO-2	Psychomotor	Guided Response (Execute)	Practical Evaluation (PECL)	1 Rubric Step
SUAS-LRT-1002-ELO-3	Psychomotor	Mechanism (Upload)	Practical Evaluation (PECL)	1 Rubric Step
SUAS-LRT-1002-ELO-4	Psychomotor	Mechanism (Capture)	Practical Evaluation (PECL)	1 Rubric Step
SUAS-LRT-1002-ELO-5	Affective	Characterization (Support)	Practical Evaluation (PECL)	1 Rubric Step
SUAS-LRT-1003-ELO-1	Cognitive	Knowledge (Identify)	Oral Examination	1 Essay/Explain Prompt
SUAS-LRT-1003-ELO-2	Psychomotor	Mechanism (Spot)	Practical Evaluation (PECL)	1 Rubric Step
SUAS-LRT-1003-ELO-3	Psychomotor	Mechanism (Modify)	Practical Evaluation (PECL)	1 Rubric Step
SUAS-LRT-1003-ELO-4	Psychomotor	Mechanism (Track)	Practical Evaluation (PECL)	1 Rubric Step
SUAS-LRT-1003-ELO-5	Affective	Valuing (Value)	Practical Evaluation (PECL)	1 Rubric Step

6.2 Comprehensive Test Items

Written Examination Items

AD-OPS-1001-ELO-1 (Attack Drone Safety)

Question 1: Prior to commencing any kinetic flight operations with an Attack Drone (AD), which component must be physically inspected to prevent an uncommanded payload release?

- A) Goggle visual input configurations
- B) The physical safety lock pin on the release servo latch (SRL)
- C) The GPS antenna polarization mast
- D) The GCS stylus calibration coordinates

Answer: B

Reference: MCRP 3-10.3 (AD Operations)

Question 2: Under which temperature threshold condition must Lithium-Polymer (LiPo) drone batteries be immediately moved to a climate-controlled storage facility to prevent cell swelling?

- A) Sustained temperatures exceeding 120°F (49°C)
- B) Sustained temperatures below 60°F (15°C)
- C) During standard 1C charging cycles
- D) When cell voltage reaches 3.85V per cell

Answer: A

Reference: System TM (Battery Care Section)

SUAS-MCRO-1001-ELO-1 (Micro-VTOL GPS Failsafes)

Question 3: What is the minimum number of active GPS satellite connections required to execute a safe autonomous launch of a Micro-VTOL (MCRO) platform without risking positional drift?

- A) 4 Satellites
- B) 6 Satellites
- C) 8 Satellites
- D) 12 Satellites

Answer: C

Reference: System TM (MCRO Operations)

SUAS-MRT-1001-ELO-1 (MRT Flight Envelopes)

Question 4: During a manual rail-launched departure of an MRT platform under high-wind conditions, the Mission Operator (MO) must delay the transition from Manual (MAN) to Altitude-Hold (ALT) mode until the aircraft has cleared which minimum altitude threshold?

- A) 50 feet AGL
- B) 100 feet AGL
- C) 150 feet AGL
- D) 200 feet AGL

Answer: B

Reference: CNAF 3710.9 (Launch Thresholds)

Oral Examination Prompts & Rubrics

AD-OPS-1001-ELO-2 (Battery Thermal Hazards)

Examiner Prompt: "Explain the physical process of thermal runaway in Lithium-Polymer (LiPo) batteries, the symptoms of impending cell failure, and the immediate safety actions required if a battery exhibits overheating during post-flight checks."

Assessment Standards (Pass/Fail):

- *Mechanism Explanation (Pass):* Correctly explains chemical gas buildup and short-circuit dynamics during runaway. *(Fail):* Fails to identify gas expansion or describes incorrect chemical dynamics.
- *Symptom Identification (Pass):* Identifies cell swelling (puffing), localized heat emission, and sweet-smelling chemical odors. *(Fail):* Misses physical puffing or temperature changes as critical symptoms.

- *Safety Action Execution (Pass)*: States that the battery must be placed in a sand-filled fire-retardant charging bag and isolated. *(Fail)*: Suggests leaving the battery in the aircraft or placing it near flammable materials.

SUAS-MRT-1002-ELO-1 (Coordinate Conversions)

Examiner Prompt: "Describe the practical steps to convert GCS map coordinate formats from Military Grid Reference System (MGRS) to Latitude/Longitude (LAT/LONG) on the RSTA laptop, and explain how a coordinate mismatch impacts combined-arms fire support coordination."

Assessment Standards (Pass/Fail):

- *Software Navigation (Pass)*: Details navigating to the RSTA GCS map configuration menu and selecting grid display properties. *(Fail)*: Cannot locate the software coordinate format transformation menu.
- *Coordinate Conversion (Pass)*: Correctly translates a sample 10-digit MGRS coordinate (e.g., 18S UT 1234 5678) into degrees-minutes-seconds. *(Fail)*: Fails to state the correct format change sequence or mathematical conversion.
- *Tactical Context (Pass)*: Explains that a mismatch leads to target-location errors, compromising friendly unit safety. *(Fail)*: Describes coordinate mismatch as an administrative error with no combat impact.

Performance Evaluation Checklist (PECL)

Test Event: SUAS-MCRO-1003 (Micro Precision & Air-Grab Recovery)

Task: Conduct Heads Up Precision Flight and Safe Recovery

Condition: Given an assembled Micro VTOL platform, GCS, a trained Visual Observer (VO), and a designated flight area with close-proximity obstacles.

Standard: Perform precision flight maneuvering near an obstacle, execute an air-grab recovery, and render the system safe without injury to personnel or damage to equipment.

Candidate Name: _____ **Rank:** _____ **Date:** _____

Evaluator Name: _____ **Result:** [] PASS [] FAIL

Step #	Performance Evaluation Standard	Objective Mapped	[P] Pass	[F] Fail
1	Evaluated wind speed using an anemometer to verify local velocity is below the 15-knot launch limit.	SUAS-MCRO-1003-ELO-5		
2	Executed pre-flight systems sweep and confirmed GCS controller pairings and battery voltage values.	SUAS-MCRO-1003-ELO-1		
3	Launched the aircraft cleanly, navigating immediate local obstacle hazards to reach a minimum clearance.	SUAS-MCRO-1003-ELO-2		
4	Maneuvered the Micro-VTOL within 2 meters of a simulated target obstacle to capture clear optical data.	SUAS-MCRO-1003-ELO-2		
5	Returned the aircraft to the landing zone, establishing a stable hover adjacent to the Visual Observer.	SUAS-MCRO-1003-ELO-3		
6	Rotated the aircraft so the onboard camera lens faced directly away from the recovery team.	SUAS-MCRO-1003-ELO-3		

Step #	Performance Evaluation Standard	Objective Mapped	[P] Pass	[F] Fail
7	Executed a physical air-grab by grasping the landing chassis firmly at arm's length from the side.	SUAS-MCRO-1003-ELO-4		
8	Initiated the dual-stick motor kill command immediately upon grasping the aircraft frame.	SUAS-MCRO-1003-ELO-4		
9	Disconnected the battery packs and packed all elements securely in temperature-controlled cases.	SUAS-MCRO-1003-ELO-5		

Critical Step Failures: Failure of steps 1, 6, 7, or 8 constitutes an immediate automatic failure of the entire evaluation due to safety violations.

6.3 Documented Program Structure

The finalized sUAS curriculum sequences the 13 T&R events into three logical instructional modules. In strict accordance with SATE instructions, no time durations are attached to lessons or modules.

Module 1: Foundations and Micro-VTOL Flight Operations

Module Description: Covers administrative, medical, and technical course prerequisites, GCS base-station setups, and fundamental quad-rotor flight physics. Focuses on close-range, low-altitude visual and screen-based flight.

Prerequisite Verification:

- BUQ-1 Course completion.
- Designation letter verified by SUAS PM.
- Airspace Restricted Operations Zone (ROZ) approval and signed spectrum request.
- NAVMED 6410/13 Medical flight clearance.

T&R Events Integrated:

- **SUAS-MCRO-1001:** Conduct Basic Heads Up Flight
- **SUAS-MCRO-1002:** Conduct Basic Heads Down Flight
- **SUAS-MCRO-1003:** Conduct Heads Up Precision Flight

Module 2: Tactical sUAS Operations (SRT, MRT, LRT)

Module Description: Progresses to complex tactical platforms. Instructs students in digital and physical hand-controller maneuvers, RSTA software path alterations, coordinate conversion, camera payload tracking, and multi-vehicle coordination.

T&R Events Integrated:

- **SUAS-IQT-1002:** Conduct Incidental Operator Qualification Training (SRT)
- **SUAS-SRT-1001:** Conduct Basic GUI Flight
- **SUAS-SRT-1002:** Conduct Basic System Hand Controller Flight
- **SUAS-IQT-1003:** Conduct Incidental Operator Qualification Training (MRT)
- **SUAS-MRT-1001:** Conduct Basic Flight

- **SUAS-MRT-1002:** Conduct Flight Mode Familiarization and MO Assisted Flight
- **SUAS-LRT-1001:** Conduct Basic Flight
- **SUAS-LRT-1002:** Conduct Flight Mode Familiarization Flight
- **SUAS-LRT-1003:** Conduct Target Acquisition Flight

Module 3: Attack Drone (AD) Employment

Module Description: The capstone module of the curriculum. Focuses on safe handling, battery safety precautions, pre-flight checklists, coordinate tracking, targeting zooms, dive-bombing runs, and physical release servo recovery steps for Attack Drones.

T&R Events Integrated:

- **AD-OPS-1001:** Perform Handling and Operations